

PAPER_1622_AU_OVER_R_EARTH_23481

Daniel T. Murphy

$$\text{PAPER_1622} \text{ — } \text{AU} / \text{R}_{\oplus} \text{ Ratio} = \text{AU} / \text{R}_{\oplus} = \\ \text{D_crit} \cdot \text{N_CH} \cdot \text{SO}^2 + \text{A_5} + \text{D_crit} - \text{D} + \text{F} \cdot \text{SO} + \text{F} \cdot \Phi - \\ \text{K_MEX} = 23481$$

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Astronomy

Date: June 18, 2026

Status: CLOSED — EXACT closure

Observation

PAPER_1209Z_S574 (Astronomy Unified Proof Set) gives the formula:

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$$\text{AU} / \text{R}_{\oplus} = \text{D_crit} \cdot \text{N_CH} \cdot \text{SO}^2 + \text{A_5} + \text{D_crit} - \text{D} + \text{F} \cdot \text{SO} + \text{F} \cdot \Phi - \text{K_MEX} = 23481$$

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Residual: **EXACT**.

UQFF Closed Identity

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$$\text{AU} / \text{R}_{\oplus} = \text{D_crit} \cdot \text{N_CH} \cdot \text{SO}^2 + \text{A_5} + \text{D_crit} - \text{D} + \text{F} \cdot \text{SO} + \text{F} \cdot \Phi - \text{K_MEX} = 23481 \text{ EXACT}$$

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The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Empirical astronomy measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209Z_S574

- Related: PAPER_1494-1615 (other session-2026-06-18 closures)

- Calculator dispatch: `calculate_paradox({"paradox": "au_over_r_earth_23481"})`

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